CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2018 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is a Final Variable? Give an example.

Definition: A variable declared with the `final` keyword becomes a constant whose value cannot be reassigned once initialized:

```
final double PI = 3.14159;
// PI = 3.14; // Compile-time error: cannot assign a value to final variable
```

Q2 (3M) What is Garbage Collection in Java?

Definition: The automatic process of finding and freeing heap memory allocated to objects that are no longer referenced by any active thread.

Q3 (3M) What is an Abstract Method?

Definition: A method declared without a body (implementation) using the `abstract` keyword, terminating with a semicolon. Must be implemented by subclasses.

Q4 (3M) What is a Layout Manager in Java AWT?

Definition: An interface/class that manages the automatic positioning and sizing of GUI components within a container window.

Q5 (3M) What is the purpose of the 'super' keyword?

Definition: References the immediate superclass object to call parent constructors (`super()`) or access hidden parent members.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) What are Packages? How are packages created in Java? Explain and give example.

Packages: Namespaces organizing classes and avoiding naming collisions. Created using the `package` keyword at the top of a source file:

```
package com.ccsu.bca;
public class Student {
    public void display() { System.out.println("BCA Student"); }
}
```

Q7 (7.5M) Explain Interface in Java with an example of Multiple Inheritance.

Explanation: Pure abstraction tool allowing classes to implement multiple interfaces simultaneously, overcoming single class inheritance limits.

Q8 (7.5M) **Explain Applet Life Cycle methods.**

Lifecycle: `init()`, `start()`, `paint(Graphics g)`, `stop()`, and `destroy()`.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain Multithreading in Java. Write a program demonstrating Thread synchronization using synchronized methods.**

Thread Synchronization: Coordinates thread access to shared mutable resources to prevent race conditions.

```
class Table {
    synchronized void printTable(int n) {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println(n * i);
            try { Thread.sleep(300); } catch(Exception e) {}
        }
    }
}
```

Q10 (15M) **What is Exception Handling? Explain Checked vs Unchecked Exceptions with try, catch, finally, and throws.**

Detailed Exception Model: Checked exceptions verified at compile-time (e.g. `IOException`) vs Unchecked exceptions occurring at runtime (e.g. `NullPointerException`).

Q11 (15M) **Explain JDBC Architecture and steps to connect Java to a database. Write a program to insert a record.**

Complete JDBC Program: Register driver, establish connection, execute SQL insert using `Statement`, and close resources.

Q12 (15M) **What is JSP? Explain its life cycle, advantages over Servlets, and scriptlet tags.**

JSP vs Servlets: Translation, compilation, and execution lifecycle; comparison of presentation layer benefits.

Q13 (15M) **Explain Event Handling in Java AWT with Delegation Event Model.**

Delegation Event Model: Event sources generate event objects delivered to registered listener interfaces.
